

Starter's Hackathon

Apopo's HeroRATs

06-07/10/2026

Intro

Why this Hackaton

You are learning a lot...

- 🔗 You learned a lot of **theory**
- 🔗 Did some **hands-on technical** courses
- 🔗 You feel like you are **mastering step-by-step the technics**

... but handling a customer is more than only technics

- 🔗 How do you organize yourselves?
- 🔗 How do you run meeting? How do you introduce yourself?
- 🔗 How do you tackle stressful moments?
- 🔗 How do you cope with an unforeseen situation?
- 🔗 How do we work together; taking notes, working on the same documents?

What is this Hackaton?

It's an opportunity to learn...

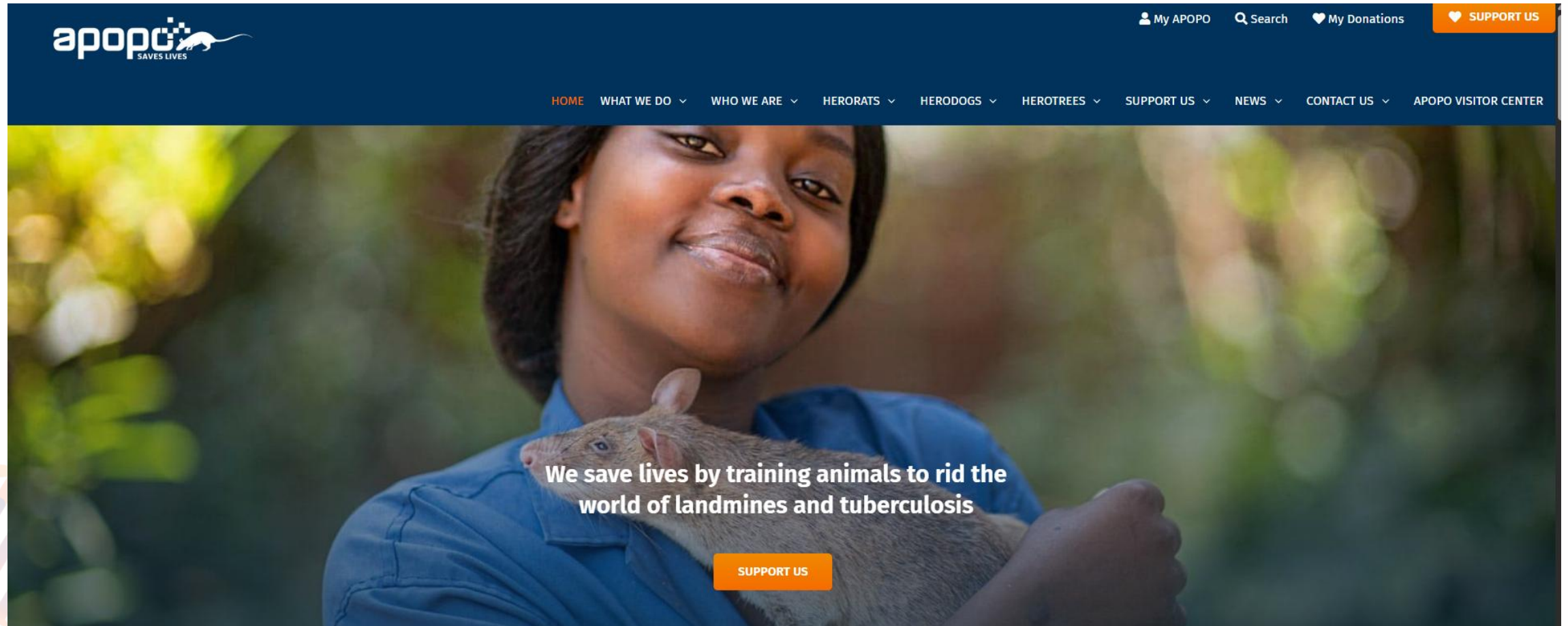
- 🔗 Learn how to **work** together
- 🔗 Learn how to **prepare** a presentation or interview
- 🔗 Learn **to (not) run** a customer interview or customer presentation
- 🔗 Learn how to **organize** yourselves
- 🔗 Learn how to get raw data into insights

= **hard & soft-skills!**

It's not meant to be...

- 🔗 A **48-hrs-no-sleep-challenge**
(Just do normal working hours)
- 🔗 A team **competition**.
Tip: Learn rather than compete!
- 🔗 The same thing year after year
(it's thus not the same dataset)

What is this hackathon about?



<https://www.youtube.com/watch?v=MHJk8ODT6es>

What is this hackathon about?

From: Charles Cuigniez <charles.cuigniez@element61.com>
To: Pieter Sterkens <pieter.sterkens@apopo.org>
Date: 12 September 2025, 16:12
Subject: RE: Follow-up on APOPO Collaboration – Deep Dive Session

Hi Pieter,

Many thanks for sharing the data and the extra info, this is very helpful! We'll review everything with the team and get back to you if we have any questions.
Just to keep you in the loop: the hackathon is scheduled for October 2nd and 3rd, with final presentations planned for the afternoon of the 3rd. We're looking forward to showcasing some innovative approaches and will keep you updated as we progress.
Thanks again and speak soon!

Kind regards,
Met vriendelijke groeten / bien à vous,
Charles Cuigniez

⇒ Mark the date and register for our [Olik Analytics Day 2025](#) on October 9th, 2025

Visit the [element61](#)-website for a list of [Events](#), [Case studies](#), [Insights](#) or our [Academy Courses](#)!



Charles Cuigniez
Director Data Science & Analysis
M [+32 494 42 87 09](tel:+32494428709)
E charles.cuigniez@element61.be



Thought-leading Analytics - Meet the most Senior Team in Belgium!
1.200+ man-years of experience in Analytics, Performance Management and Data Science



From: Pieter Sterkens <pieter.sterkens@apopo.org>
To: Charles Cuigniez <charles.cuigniez@element61.com>
Date: 12 September 2025, 15:47
Subject: RE: Follow-up on APOPO Collaboration – Deep Dive Session

Hi Charles,

As discussed, I'm sending over a sample of the data and some extra information about the variables and our current data collection process. Please let me know if you need any additional context or if there are specific variables you'd like us to include for the hackathon.
Some of the behavioural patterns in the data, especially around sniff duration thresholds, might be worth a closer look. They've raised interesting questions internally about how we interpret subtle signals and edge cases.
I'm curious to see what your teams might uncover when exploring those areas.

BR,

Pieter Sterkens
Head of Training and Research



Who is who?

Hackathon main stakeholders



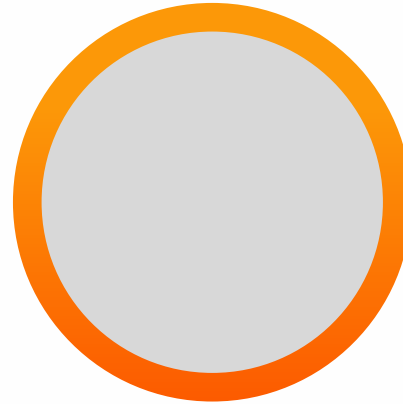
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Project Manager

Main internal contact for this project

Your partner-in-crime

sharing experience & expertise



Who is who?

Hackathon main stakeholders



Apopo

Head of Training & Research

Has needs & concerns
Wants to make a good impression
for the board

sharing experience & expertise



Apopo

Senior Data Analyst

Knows the data & tools
Rather technical background
Has needs & concerns



Apopo introduction



Hackathon Lab

APOPO

■ APOPO

1. A non-profit organization
2. Develops and deploys innovative scent detection technology
3. Addressing urgent global health (TB) and humanitarian challenges (landmines, Search and Rescue, Illegal Wildlife Traffic)
4. One of its most impactful programs focuses on the detection of tuberculosis (TB)

■ Tuberculosis key facts

1. Worldwide, TB has probably returned to being the world's leading cause of death from a single infectious agent.

- A total of 1.25 million people died from TB in 2023
- An estimated 10.8 million people fell ill with TB worldwide in 2023
- TB is present in all countries and age groups. TB is curable and preventable.
- Global efforts to combat TB have saved an estimated 79 million lives since the year 2000.

2. Only about 2 in 5 people with Multidrug-resistant TB (MDR-TB) has access to treatment.

- US\$ 22 billion is needed annually for, TB prevention diagnosis, treatment and care to achieve the global target by 2027 agreed at the 2023 UN high level-meeting on TB.
- Ending TB epidemic by 2030 is among the health targets of the United Nations Sustainable Development Goals (SDGs).

■ Tuberculosis Detection @ APOPO

1. APOPO trains African giant pouched rats, known as HeroRATs, sniffing out TB in human sputum samples
2. These rats are fast, accurate, and cost-effective
3. Ability to screen hundred of samples in a fraction of the time traditional labs require
4. Used as second-line screening, working alongside partner clinics to identify TB-positive cases that might otherwise be missed, especially in low-resource settings
5. By rapidly identifying additional TB cases, APOPO helps improve detection rates, reduce the spread of the disease, and accelerate treatment, contributing directly to better public health outcomes and the global fight against TB



■ **Supporting APOPO**

1. **APOPO is actively seeking R&D collaborations with TB diagnostics and pharmaceutical partners to expand the impact of its technologies, including their data backbone.**
2. **We are also building access and scale-up partnerships in TB-endemic countries, working closely with local health systems and stakeholders.**
3. **We welcome opportunities to co-develop validation protocols, pilot programs, or innovative solutions, offering partners a unique platform for testing, implementation, and entry into high-burden markets.**
4. **In parallel, APOPO has considerable amount of data, which could be analysed and unlock new opportunities**

■ Proposed operational project: Sniff times

Provide **near real-time direction** on which samples to consider suspect of containing TB based on how long the individual rat sniffed the sample.

Currently, rats are rewarded for correctly indicating a TB+ sample when they hold their nose over (sniff) the sample for a predetermined duration. This so-called "**indication threshold**" varies from rat-to-rat.

Apopo rats currently achieve a 95% sniff-accuracy. To get FDA approved, **98% accuracy needs to be reached**.



Apopo has a lot of historical data. We want to leverage your expertise in data & analytics to extract key insights from the data and help us define data-driven approaches to **bridge the 3% gap**.



■ Proposed technical project: Data backbone

Beyond the sniff times: give APOPO **trusted, audit-proof data** and **fast, trustworthy answers** for researchers, auditors, donors and the board.

T1 · DATA FOUNDATION

LIMS migration without losing trust

- Migrate to the new LIMS while keeping one trusted, audit-proof version of the study data

KPI 100% of records reconciled, identical KPIs before and after, zero study downtime

T2 · SELF-SERVICE

Data & audit copilot

- Let researchers and auditors query APOPO's data themselves and get answers they can trust

KPI time-to-answer from days to minutes, % correct on an agreed question set, analyst hours freed

T3 · REPORTING

Automated impact & audit reporting

- Consistent, audience-specific reports in hours instead of days

KPI report cycle from days to hours, one number per KPI across audiences, cost per case available

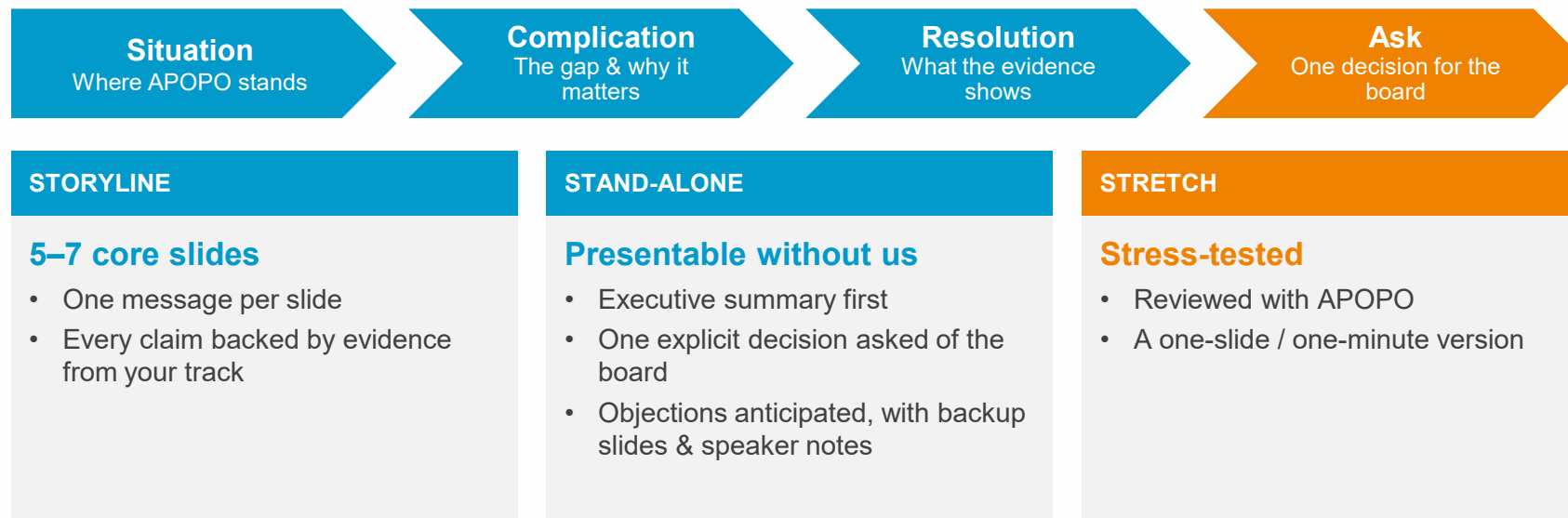




sharing experience & expertise

■ Project Goal: Board-ready pitch

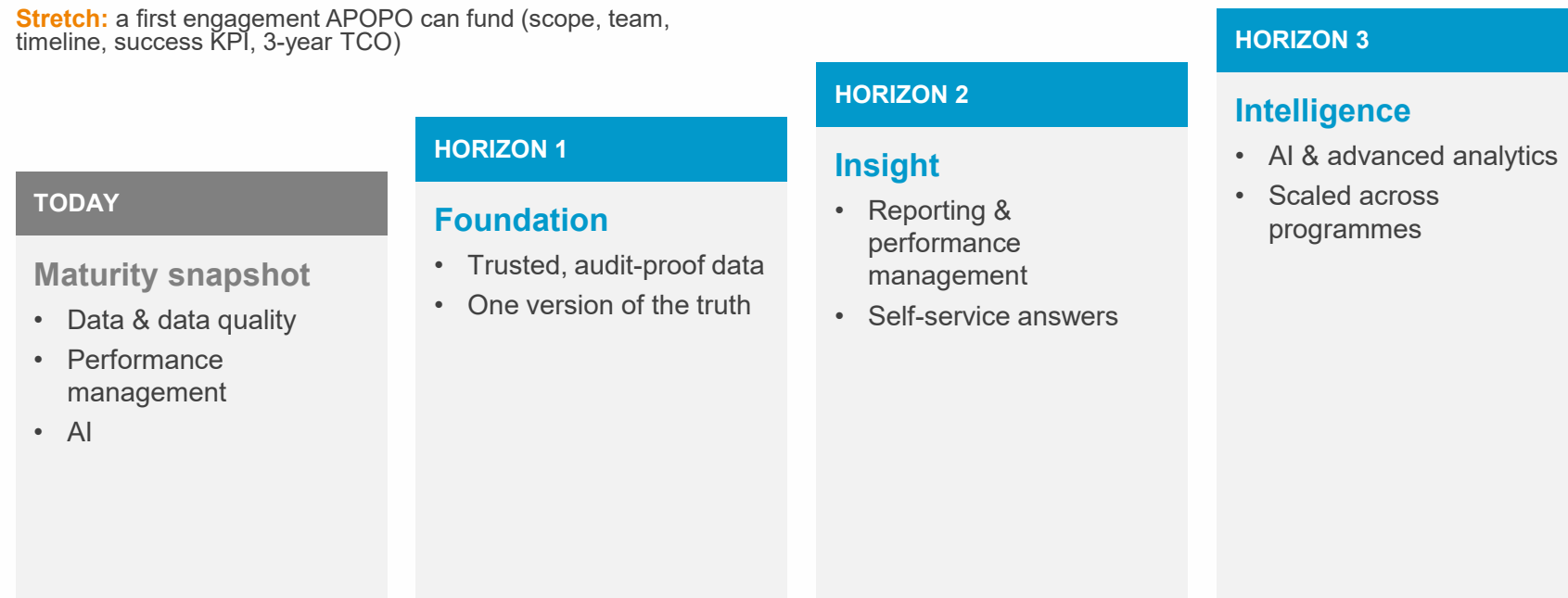
APOPO's Head of Training & Research needs to present to the board. **Turn your evidence into a decision the board can take**, in a storyline he can present himself.



■ Project Goal: Roadmap

The board wants to know **where APOPO stands** on data, reporting and AI, and **what it would cost to get there**. Give them a realistic path forward, with a price tag.

Stretch: a first engagement APOPO can fund (scope, team, timeline, success KPI, 3-year TCO)



Per horizon: one-off (CAPEX) and yearly running (OPEX) costs, as ranges with visible assumptions

Practicalities

Available data



APOPO Hackathon 2026 – Group Data Guide.docx

1x Data catalog

Data Variable	Description
SESSION_DATE	Date the evaluation session was conducted (i.e., when the rats
PROGRAM	This dataset includes results from rats working at both our Dar e Salaam ("DAR") and Morogoro ("MORO") facilities. NOTE: DAR
ID_EVALUATION_SESSION	Unique ID for each evaluation session. Rats typically participate in a single evaluation session per day with each session
ID_SAMPLE	Unique sample ID (note: patients may provide more than 1
RAT_NAME	Name of the rat evaluating the sample
RUN	Samples are presented to the rats in bars of 10 labeled alphabetically. RUN A contains samples 1-10, RUN B contains
HOLE	This number corresponds to the hole position with the sample bars (RUN). The first sample a rat encounters in an evaluation session is A1, the 10th sample is A10, 11th is B1... Combining RUN
HIT	Defined as a continuous sniff (nose insertion into cage hole containing the sample) equal to or exceeding approximately 3-s (as estimated by the rat handler). TRUE means the rat "indicated" the sample (held its nose in the hole for at least 3-s). FALSE
STATUS_BLINDPOS	A sample found to be TB-positive from the DOTS clinic but which is coded as TB-negative during the evaluation session. Blind samples ensure that rat behavior is driven by the smell of the sample and not inadvertent cueing from the handler. They also teach that rat that they will occasionally encounter TB-positive



AutomatedCage_Sept2023-24_raw data_with PatientID.xlsx

1x Data record

SESSION_DATE	RAT_NAME	SniffTime	SniffThreshold
15-Mar-19	Daudi	0	2500
15-Mar-19	Daudi	3831	2500
15-Mar-19	Daudi	0	2500
15-Mar-19	Daudi	0	2500
15-Mar-19	Daudi	0	2500
15-Mar-19	Daudi	0	2500
15-Mar-19	Daudi	0	2500
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15-Mar-19	Daudi	0	2500

Timing

Tuesday	
9 AM – 10 AM	Hackathon briefing - Share mail chain & calendar invite (Monday interviews, Tuesday noon sync & final presentation) - Set-up groups
10 AM – 12 PM	Interviews Apopo - 20 min / interview
Wednesday	
9 AM – 11 AM	Standup Parc
3 PM – 5 PM	Final presentation

Team composition

Group 1

Douwe Pauwels
Sena Oztarhan
Wout Mostrey
Alexandre Paice

Group 2

Matheus Ramos
Artemis Xhuri
Senne Bosschem

Group 4

Alexis Vendrix
Joachim Vande Ginste
Wout Belmans

Group 3

Michael Morozov
Stefanos Panagoulas
Paulina Rocha
Sebbe Blancke

Rejection threshold

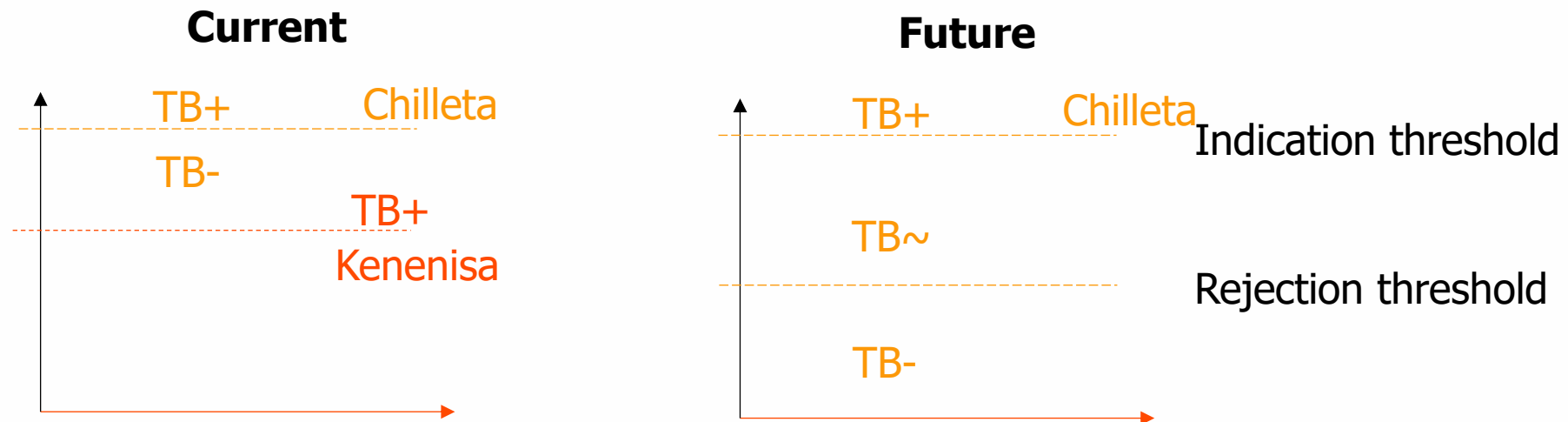
■ Indication threshold

1. This is the minimum sniffing duration (in milliseconds) that a specific rat must spend on a sample for it to be flagged as suspect of containing TB.
2. It is individualized per rat — e.g., Ella 1,000 ms, Kenenisa 2,250 ms (typical values in the 2023–24 data).
3. If the rat's sniff reaches its threshold, the system treats this as a TB+ indication, and the sample is sent for lab confirmation.
4. It is used to reward the rat during training and actual diagnosis.

■ Rejection threshold

1. This is a lower sniffing duration limit that may still indicate suspicion, but not strong enough to reach the full indication threshold.
2. Also customized per rat and potentially adapted per day/session depending on the rat's performance.
3. If the rat sniffs longer than usual, but not long enough to reach the indication threshold, this behavior could still suggest some uncertainty or low-level suspicion.
4. Samples in this zone might otherwise be dismissed, but the rejection threshold could flag them for additional review, potentially increasing sensitivity (i.e., catching more true TB+ cases).

■ Rejection threshold



Above indication – TB positive

Between indication and Rejection – Uncertain, follow up

Below Rejection – TB Negative

Identify the rat specific (condition specific) rejection threshold

Toolkit: the data

The data pack: two files from APOPO

File	What it is	Grain
Data/APOPO Hackathon 2026 – Group Data Guide.docx	How to read the main dataset: all 34 columns, bacterial level codes, and how APOPO defines sensitivity, specificity, false alarm and new case	Reference
Data/AutomatedCage_Sept2023-24_raw data_with PatientID.xlsb	Main dataset: rat evaluations Sep 2023 – Sep 2024 (15 rats, 645 sessions, 12,809 samples) with sniff time, thresholds, indication (HIT), reward, presentation order, clinic and lab results, patient ID	One row per rat per sample per session
Tracks/O4... · Tracks/T1...	Track-specific extras: the follow-up tracker (O4) and the new LIMS export (T1)	See track slides

The data is shared **as received** from APOPO. It isn't cleaned.